

Thanathorn (Tammy) Sukprasert

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Research Focus

Systems researcher building **GPU and CPU power management infrastructure** for AI datacenters, spanning NVML-based power capping, DVFS, utilization-aware scheduling, and power-aware cluster orchestration under grid demand response constraints. I also study the limits of carbon-aware scheduling and renewable-native system design to make AI workloads reliable under real-world energy scarcity.

Education

University of Massachusetts Amherst

Amherst, MA

MS/PhD in Computer Science

2022–Present

Advisors: Prof. Prashant Shenoy, Prof. David Irwin

Relevant Coursework: Distributed & Operating Systems, Computer Architecture, Database Systems, Algorithms

University of Massachusetts Amherst

Amherst, MA

B.S. in Computer Engineering, *Summa Cum Laude*, GPA: 3.97/4.00

2018–2022

Relevant Coursework: System Software Design, Computer Networking, Communication Systems, Design Project

Experience

Dolby Laboratories. Research Intern

Sunnyvale, CA

End-to-End Measurement of Low-Latency AR/VR Streaming Systems

Summer 2024

- Modified the ALVR client codebase on **Meta Quest** to intercept a button-press stimulus and capture the exact frame timestamp when the corresponding red pixel appeared on the display, enabling ground-truth motion-to-photon latency measurement on the device.
- Deployed client-side logging on the headset to record per-frame timestamps, isolating latency contributions across the server encoding, network transit, and client decode/render stages.
- Built a **tc-netem**-based testbed to sweep network latency, jitter, and packet loss; traced a **2× gap** between ALVR's server-reported and true end-to-end motion-to-photon latency.

University of Massachusetts Amherst. Graduate Research Assistant

2022–Present

PowerRanger: GPU & CPU Power Management for Production Datacenters

- Implemented **GPU-level power controls** via NVML: dynamic power capping, frequency/voltage scaling (DVFS), and utilization-aware scheduling to enforce fine-grained per-GPU power budgets across AI/ML workloads.
- Built **CPU power management** using cgroups, cpuset, and RAPL-based DVFS within an LXD cluster manager, enabling coordinated CPU+GPU power enforcement across container workloads.
- Developed global and local orchestration policies that translate raw grid DR signals into workload control actions (dynamic power capping vs. workload revocation), validated on a physical testbed instrumented with RAPL, IPMI, and PDU telemetry.
- Building toward **production datacenter deployment**: prototyped the control plane on real cluster hardware, with active work on hardening enforcement reliability and integration with datacenter power telemetry.

Renewable-Powered Edge Site Selection

- Investigated whether latency-sensitive distributed systems can operate entirely on intermittent renewable energy.
- Formulated site selection as a **mean-variance optimization** problem using Modern Portfolio Theory: minimize variance of renewable energy supply across selected edge sites subject to latency and availability constraints, capturing spatial diversification benefits across solar and wind assets.
- Replayed solar and wind generation traces on a physical edge testbed to validate availability gains from the optimized site selection under real renewable intermittency.

Characterizing the Limits of Carbon-Aware Cloud Scheduling

- Challenged the assumption that temporal and spatial workload shifting alone can deliver large carbon reductions.
- Designed experiments using real carbon-intensity signals and workload traces from AWS, Azure, and GCP.
- Analyzed **123 global regions (2020–2022)** under realistic datacenter capacity constraints.
- Showed that while idealized migration enables up to **96%** emission reduction, practical constraints limit achievable global reductions to **~52%**.

Comparing Average vs. Marginal Carbon Signals in Cloud Optimization

- Evaluated 65 global regions and identified weak correlations between average and marginal carbon-intensity signals.
- Demonstrated that optimizing for one signal can unintentionally increase emissions under the other.

Publications

Walid A. Hanafy, **Thanathorn Sukprasert**, Abel Souza, David Irwin, and Prashant Shenoy. *Untangling the Carbon-Cost Tradeoffs and Stampede Effect Challenges in Cloud Computing*. IEEE Transactions on Computers, Feb. 2026.

Thanathorn Sukprasert, Abel Souza, Noman Bashir, David Irwin, and Prashant Shenoy. *On the Limitations of Carbon-Aware Temporal and Spatial Workload Shifting in the Cloud*. Proceedings of the Nineteenth European Conference on Computer Systems (EuroSys). Athens, Greece. April 2024. Coupled with a poster.

Thanathorn Sukprasert, Noman Bashir, Abel Souza, David Irwin, and Prashant Shenoy. *On the Implications of Choosing Average versus Marginal Carbon Intensity Signals on Carbon-aware Optimizations*. Proceedings of the 15th ACM International Conference on Future and Sustainable Energy Systems (e-Energy). Singapore. June 2024. Short paper.

Thanathorn Sukprasert. *How the Choice of Carbon Signal Impacts Carbon-Aware Scheduling Decisions*. EuroSys Doctoral Workshop (EuroDW). Athens, Greece. April 2024.

Thanathorn Sukprasert, Abel Souza, Noman Bashir, David Irwin, and Prashant Shenoy. *Spatiotemporal Carbon-aware Scheduling in the Cloud: Limits and Benefits*. Companion Proceedings of the 14th ACM International Conference on Future Energy Systems (e-Energy Companion). Orlando, Florida. June 2023.

Technical Skills

Power Management: NVML (GPU power capping), GPU DVFS, RAPL, IPMI, ePDU, cgroups, cpuset, CPU DVFS, LXD

Programming: Python, C/C++, Bash

Systems: Linux internals, cgroups, cpuset, DVFS, LXD, Docker, Spark

Cloud & Infra: GCP, AWS, Azure, HAProxy

Measurement: NVML, RAPL, IPMI, ePDU, latency tracing, tcpdump, tc-netem

Data & Analysis: NumPy, Pandas, SciPy, scikit-learn

Research Themes: GPU/CPU power management, power-aware cluster orchestration, carbon-aware scheduling, renewable-powered infrastructure

Awards and Honors

Krithi Ramamritham Computer Science Scholarship 2026

UMass Chancellor's Scholarships (all 8 semesters, \$48,000)

Undergraduate Dean's List (all 8 semesters)

Eta Kappa Nu (IEEE-HKN), junior-year inductee

Mentorship and Teaching

Mentorship: Advised 15+ undergraduates across 5 URV projects (2023–2026) and 1 NSF REU student (MAGICC 2025).

Teaching: Distributed & Operating Systems (×2), Computer Systems Principles, Intro to Python, Mobile Health Sensing; Undergraduate Instructor (Spring 2022).

Service

PhD Research School: Advances in Learned Systems, Learning Directed Operating System Expedition (LDOS), Attendee

OSDI '26 Artifact Evaluation Committee, Evaluator

EuroSys 2026 Shadow PC (2025)

UMass CS PhD Applicant Support Program committee member (2025)

UMass CS MS Application Reviewer (2025)

UMass CS PhD Application Reviewer (2024)

UMass Turing Summer Program (2023, 2025)

UMass Summer Engineering Institute Mentor (2022)

Podcast Guest: Disseminate CS Research Podcast; Environmental Variables Podcast

PhD Peer Mentor (2024–2025)